

Early Prediction of Prolonged ICU Stay in Lung-Cancer Patients Using First-24-Hour eICU Data

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Summary

This study assessed whether information available at intensive care unit (ICU) admission and during the first 24 hours could predict an ICU stay longer than five days among adults with primary lung cancer. A retrospective prediction study was performed using the eICU Collaborative Research Database v2.0. Lung-cancer stays were identified from patient, diagnosis, and admission-diagnosis records; one eligible ICU stay per patient was retained. Patients whose ICU stay ended within 24 hours were excluded so that all predictors preceded the prediction point. The final landmark cohort contained 1,389 stays, including 286 prolonged stays (20.6%). Demographic and admission information, first-day laboratory results, vital signs, structured medical history, and APACHE variables were transformed into patient-level features. A stratified 80:20 split produced a development set of 1,111 stays and an independent test set of 278 stays. Logistic regression, random forest, and gradient-boosting models were compared using five-fold cross-validation. The final sigmoid-calibrated L1 logistic regression achieved a test ROC-AUC of 0.724, average precision of 0.412, sensitivity of 63.2%, specificity of 69.2%, negative predictive value of 87.9%, and Brier score of 0.146. It correctly identified 36 of 57 prolonged stays. Mechanical ventilation, intubation, heart rate, oxygen saturation, admission context, respiratory rate, and selected laboratory and history variables contributed most. First-day data therefore supported moderate early risk stratification, but external and prospective validation is required before clinical use.

Introduction

Prolonged ICU stays are clinically and operationally important because they often reflect complex treatment needs, high resource use, bed-capacity pressure, and difficult discharge planning. Earlier identification of patients at increased risk could support monitoring priorities, staffing, and resource allocation, although prediction must complement rather than replace clinical judgement. Electronic health records provide repeated measurements, diagnoses, treatments, and severity indicators for this purpose, but critical-care data are irregularly sampled, heterogeneous, and frequently incomplete (Johnson et al., 2016).

ICU length of stay is especially difficult to forecast because the final duration depends not only on admission physiology but also on later complications, treatment response, hospital workflow, and discharge practices. Benchmark studies have shown that length-of-stay prediction is more challenging than several other ICU tasks (Harutyunyan et al., 2019). Nevertheless, early estimates remain valuable because even moderate risk separation may help clinicians and managers focus attention on patients more likely to require prolonged critical care.

The eICU Collaborative Research Database is a deidentified, multicentre resource containing high-granularity records for more than 200,000 ICU admissions from 208 United States hospitals in 2014-2015 (Pollard et al., 2018, 2019). Its structured vital signs, laboratory results, diagnoses, treatments, medical-history records, and APACHE severity variables support reproducible prediction research. Lung-cancer patients are a clinically relevant subgroup because respiratory failure, mechanical ventilation, metastatic disease, and acute severity are prominent markers of adverse critical-care outcomes (Elkhapery et al., 2025).

The research question was: how accurately can an ICU stay longer than five days be predicted using information recorded at admission and during the first 24 hours? The primary objective was to develop and internally validate an interpretable early-risk model. Secondary objectives were to quantify the added value of APACHE information, evaluate discrimination and probability calibration, identify clinically understandable operating characteristics, and determine which variables contributed most strongly to prediction.

Methodology

Research design and data source

A retrospective observational prediction design was applied to eICU-CRD v2.0. The database is distributed as relational tables and contains deidentified records from a multicentre tele-ICU programme. All row-level data were processed locally under the applicable data-use agreement; only aggregate quality-control outputs and figures were used for reporting. The analysis unit was one ICU stay, linked across tables by patientunitstayid.

Cohort construction

Adult ICU stays with evidence of primary lung cancer were identified from diagnosis, admission-diagnosis, and patient records using ICD-9 code 162.*, ICD-10 code C34.*, structured hierarchy terms, and the exact admission diagnosis “Thoracotomy for lung cancer.” Records indicating secondary lung malignancy, mesothelioma, biopsy-pending status, isolated malignant pleural effusion, or unconfirmed thoracic surgery were excluded. One eligible ICU stay per patient was retained. The source cohort contained 1,860 stays. Because predictors were defined over the first 24 hours, 471 stays ending within 24 hours were excluded, leaving a 24-hour landmark cohort of 1,389 stays.

Outcome definition

Prolonged ICU stay was defined as a unit-discharge offset greater than 7,200 minutes, equivalent to more than five days. The five-day threshold selected a clinically meaningful high-resource subgroup while retaining enough positive examples for modelling. The target was calculated from the discharge offset but all discharge, mortality, identifier, and post-landmark fields were excluded from the predictor matrix.

Predictors and feature engineering

Predictor blocks included demographics and admission characteristics; first-day laboratory measurements; periodic and aperiodic vital signs; structured past medical history; APACHE prediction variables and acute physiology variables; and APACHE severity scores. Repeated laboratory and vital-sign measurements were summarised using first value, median, minimum, maximum, standard deviation, and measurement count where appropriate. Only information documented by the 24-hour landmark was eligible. APACHE-predicted ICU length of stay was retained only as a benchmark and never used as a predictor.

Preprocessing, modelling, and validation

The patient-level modelling table contained 278 columns, including 222 candidate predictors. A stratified 80:20 split produced 1,111 development stays and 278 untouched test stays with no patient overlap. Numeric variables were median-imputed; categorical variables were imputed and one-hot encoded; and numeric scaling was applied for logistic regression. All learned transformations occurred inside scikit-learn pipelines and cross-validation folds. Clinical-only and clinical-plus-APACHE feature sets were compared using five-fold stratified cross-validation with logistic regression, random forest, and histogram gradient boosting. Hyperparameters were tuned by randomised search using average precision. The selected logistic and random-forest models were sigmoid-calibrated. A probability cut-

off was locked from out-of-fold development predictions by maximising specificity while maintaining at least 70% sensitivity. Final performance was evaluated once on the sealed test set with 2,000-sample bootstrap confidence intervals.

From the source cohort to a fair final test

Patients had to remain in the ICU beyond 24 hours so all predictors were available before prediction

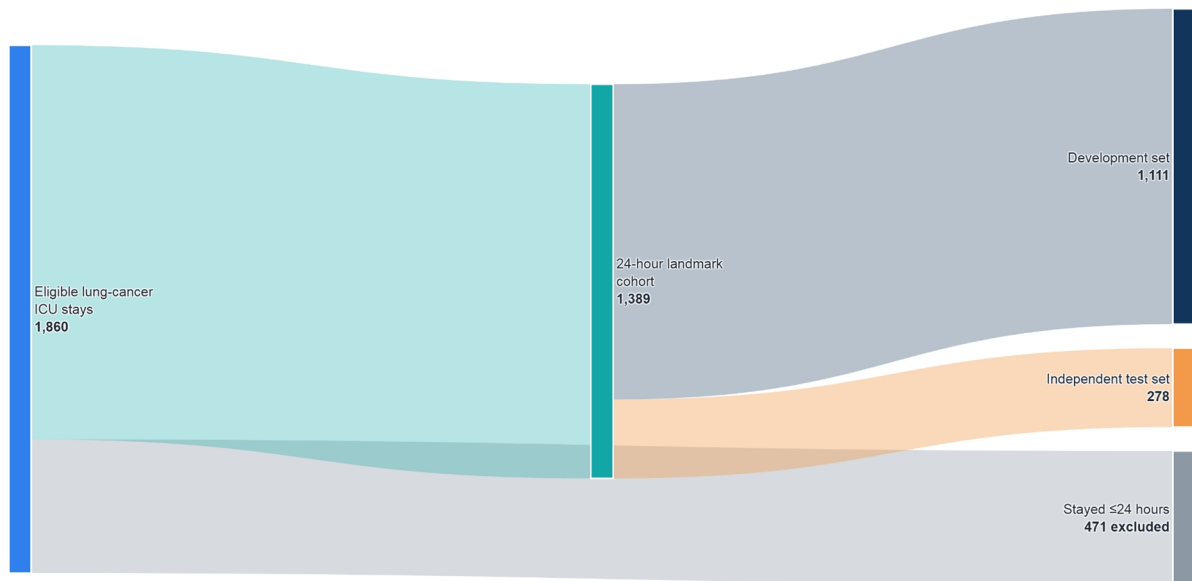


Figure 1. Cohort construction, 24-hour landmark eligibility, and stratified development/test split.

Tools and reproducibility

Data extraction and quality control were performed locally with MySQL, Python, pandas, NumPy, and Jupyter Notebook. Scikit-learn was used for preprocessing, cross-validation, model fitting, calibration, threshold assessment, and permutation importance. Plotly was used to create presentation-ready figures, and aggregate outputs were saved as versioned CSV files. Patient-level feature matrices, splits, and predictions remained in the private local project directory.

The prediction was made after the first 24 hours

A clear timeline prevents future information from leaking into the prediction

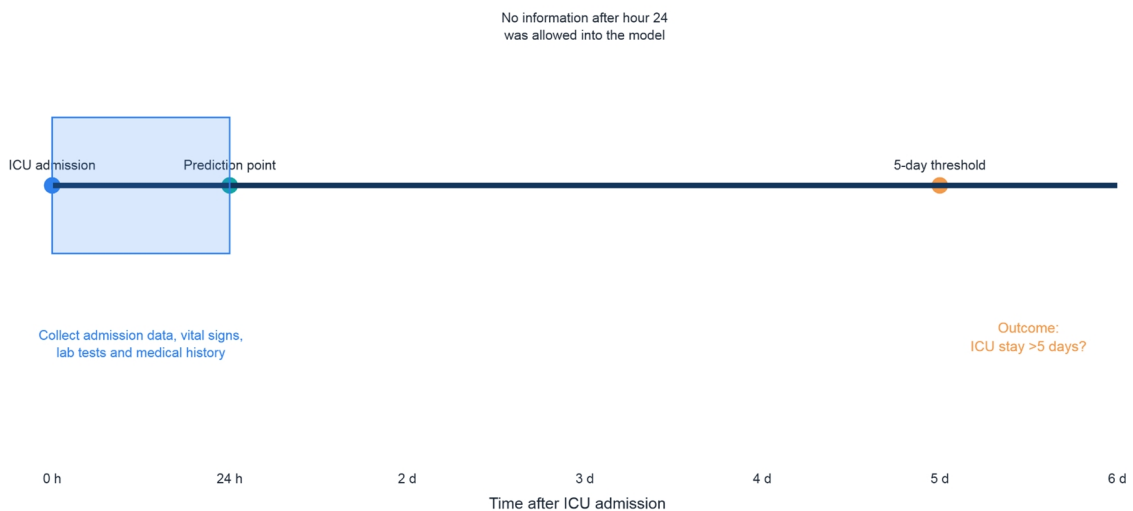


Figure 2. Prediction landmark: only admission and first-24-hour information was allowed into the model.

Table 1. Availability of major first-day predictor blocks

Predictor block	Patients available	Coverage
Vital signs	1,370	98.63%
Assessable medical history	1,360	97.91%
Laboratory data	1,349	97.12%
APACHE physiology	1,335	96.11%
APACHE admission variables	1,335	96.11%
APACHE result / severity score	1,248	89.85%

Results

Cohort and cross-validation findings

The landmark cohort comprised 1,389 unique ICU stays; 286 (20.59%) exceeded five days. The development set contained 1,111 stays and 229 prolonged stays (20.61%), while the test set contained 278 stays and 57 prolonged stays (20.50%). No all-missing predictors remained in either principal feature set, and independent leakage audits removed hospital-mortality and ICU-mortality outcomes before modelling.

In five-fold development cross-validation, adding APACHE information improved average precision and ROC-AUC for logistic regression, random forest, and gradient boosting. The strongest baseline combination was the clinical-plus-APACHE random forest, with average precision 0.396 ± 0.050 and ROC-AUC 0.705 ± 0.041 . After tuning, pooled out-of-fold average precision was 0.408 for logistic regression and 0.388 for random forest; the corresponding ROC-AUC values were 0.715 and 0.718. Sigmoid calibration reduced the logistic model Brier score from 0.210 to 0.148 and aligned its average predicted risk with development prevalence.

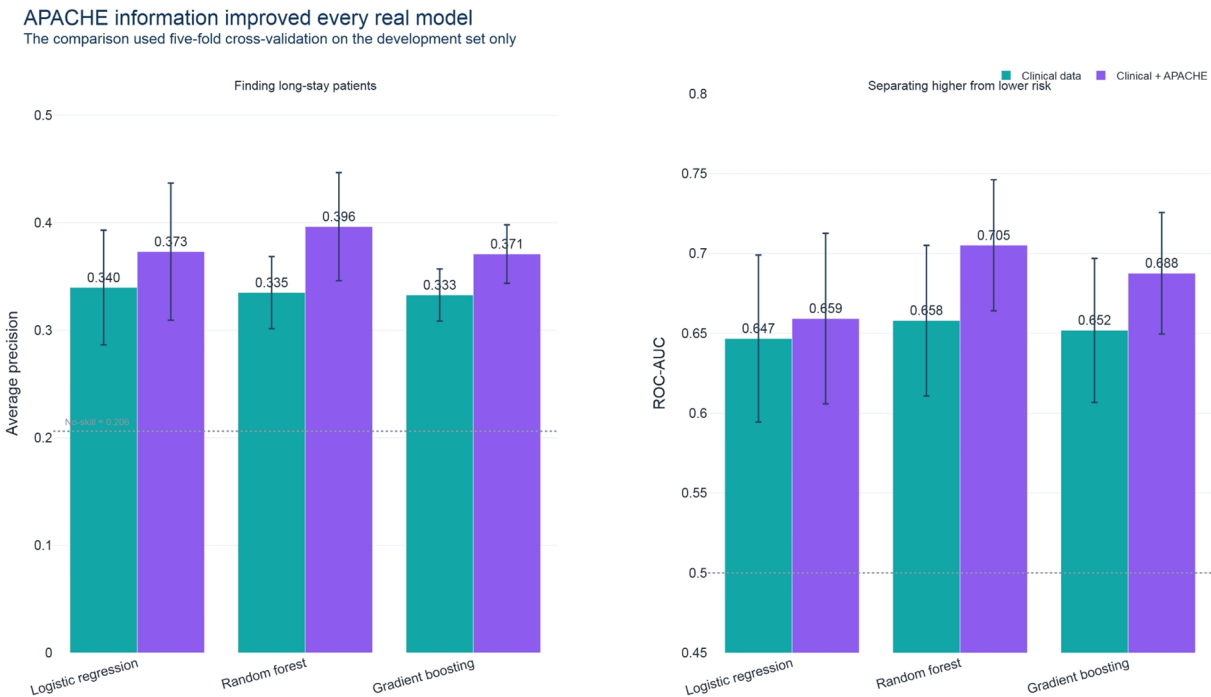


Figure 3. Five-fold development cross-validation: APACHE information improved all evaluated real models.

Independent test performance

The calibrated L1 logistic regression was selected and its operating threshold was locked before test evaluation. On the independent test set, it achieved ROC-AUC 0.724 (95% CI 0.649-0.798), average precision 0.412 (0.306-0.549), and Brier score 0.146 (0.119-0.172). At the locked threshold of 0.195, the model identified 36 of 57 prolonged stays. The confusion matrix contained 153 true negatives, 68 false positives, 21 false negatives, and 36 true positives. The calibrated random forest produced the same ROC-AUC of 0.724 but lower average precision of 0.381.

Table 2. Independent test performance of the primary model

Metric	Estimate	Bootstrap 95% CI
ROC-AUC	0.724	0.649-0.798
Average precision	0.412	0.306-0.549
Sensitivity	63.2%	50.0%-76.2%
Specificity	69.2%	63.3%-74.8%
Precision	34.6%	25.5%-43.5%
Negative predictive value	87.9%	82.9%-92.7%
F1 score	0.447	0.346-0.538
Balanced accuracy	0.662	0.594-0.731
Brier score	0.146	0.119-0.172

Discussion

The first 24 hours contained a reproducible but incomplete signal for prolonged ICU stay. A ROC-AUC of 0.724 indicates moderate ranking ability, while average precision of 0.412 was approximately twice the 20.5% test prevalence. The model was more reliable for identifying lower-risk patients than confirming prolonged stay: negative predictive value was 87.9%, whereas only 34.6% of high-risk classifications were true prolonged stays. Its practical role is therefore early risk stratification and prioritisation for review, not an automated decision about admission, discharge, treatment, or resource restriction.

APACHE information improved every evaluated model, supporting the predictive importance of acute severity and admission context. APACHE IV was developed for severity assessment and ICU benchmarking, including length-of-stay estimation (Zimmerman et al., 2006). Mechanical ventilation, intubation, heart rate, oxygen saturation, hospital-to-ICU timing, respiratory rate, admission diagnosis, and previous venous thromboembolism were among the strongest permutation signals. Their prominence is clinically plausible and broadly consistent with lung-cancer ICU evidence emphasising respiratory support and severity of illness (Elkhapery et al., 2025). These are predictive associations and should not be interpreted as causal effects.

The result is comparable with earlier first-day ICU length-of-stay studies, although populations and thresholds differ. Alghatani et al. (2021) reported approximately 65% accuracy for a vital-sign-based binary model. A recent MIMIC-IV study reported AUROC 0.732 using structured first-day data without delayed diagnostic embeddings, close to the present 0.724 (She & Wood-Doughty, 2026). More complex temporal models may improve detailed remaining-length predictions but require larger datasets, more computation, and less transparent interpretation (Rocheteau et al., 2021). Important limitations of the current study include retrospective design, structured-code cohort identification, heterogeneous hospital documentation, internal rather than external validation, a relatively small test set, and selection of

patients remaining in the ICU beyond 24 hours. Future work should evaluate hospital-grouped, temporal, and external validation; subgroup fairness; dynamic predictions after 24 hours; and the incremental value of notes and treatment trajectories.

Conclusion

Admission information and first-24-hour clinical data predicted an ICU stay longer than five days with moderate accuracy in critically ill lung-cancer patients. The calibrated logistic model combined discrimination, calibration, and interpretability, but missed more than one-third of prolonged stays and produced substantial false alarms. It may support research-oriented early warning and planning, but external and prospective validation are required before clinical deployment.

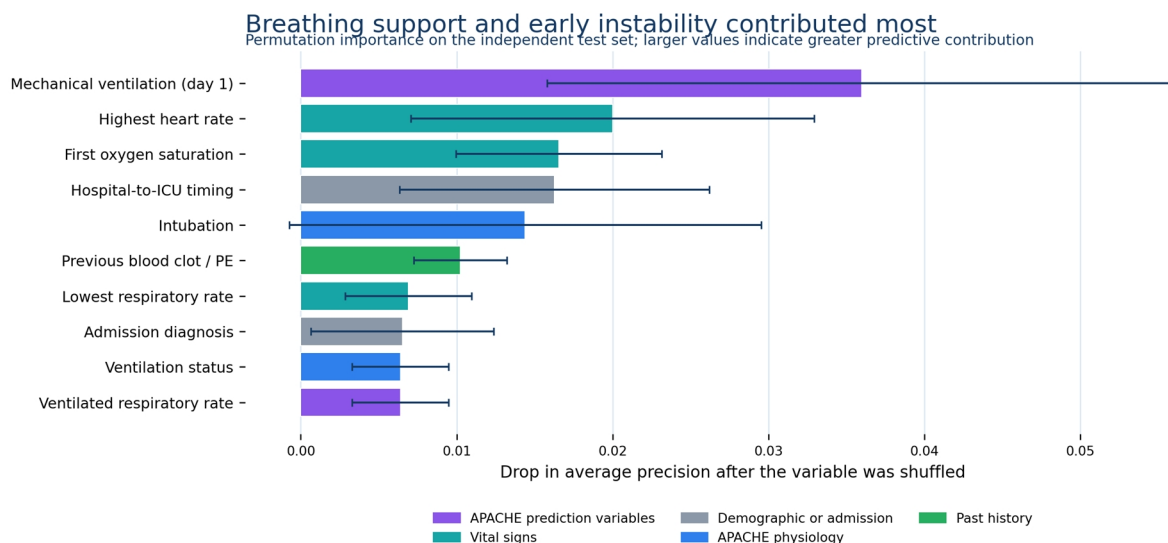


Figure 4. Ten variables with the largest test-set permutation importance for the primary model.

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